# Spectral–Temporal–Spatial Feature Optimization for *Dioscorea Polystachya* Turczaninow Classification Using Time Series Sentinel-2 Data

Zhulin Chen, Tingting Shi, Haiying Jiang, Bo Yuan, Yuran Cui, Shijiao Qiao, and Kun Jia

***Abstract*— *Dioscorea polystachya* Turczaninow is one of the most famous traditional Chinese Materia Medica. However, there is a lack of large-scale classification method, which is crucial for its growth status monitoring and yield estimation. This study proposed a reliable *D. polystachya* Turczaninow classification model based on spectral–temporal–spatial feature optimization using time series Sentinel-2 data. First, 16 monotemporal classification models were developed using five vegetation indices (VIs) and random forest (RF) algorithm. Then, temporal feature optimization was conducted by identifying the most effective time phase combinations based on Sentinel-2 time series normalized difference vegetation index (NDVI) data, Gaussian mixture modeling algorithm, and *F*1 score of *D. polystachya* Turczaninow in each monotemporal model. Next, spectral features were optimized by replacing NDVI with the optimal VI corresponding to each time phase, thus constructing a multi-VI-based time series dataset. Finally, the spatial feature optimization was conducted using the 3-D convolutional neural network (3-D CNN) algorithm and the multi-VI-based time series Sentinel-2 data. Following the comprehensive feature optimization, the final *D. polystachya* Turczaninow classification model was determined. The results found that Sentinel-2 data acquired during the rhizome enlargement stage played a crucial role in classifying the *D. polystachya* Turczaninow. By using the optimized features, the classification model achieved the *D. polystachya* Turczaninow *F*1 score of 95.00%, which improved by 11.49% compared to only using the time series NDVI data. This spectral, temporal, and spatial feature optimization method also has the potential to the development of large-scale, dynamic, and accurate mapping for other crops.**

***Index Terms*— Classification, *Dioscorea polystachya* Turczaninow, feature optimization, remote sensing time series, Sentinel-2.**

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## I. Introduction

*DIOSCOREA polystachya* Turczaninow is one of the most widely used traditional Chinese Materia Medica [1]. Boasting a wealth of nutritional content and medicinal compounds, *D. polystachya* Turczaninow is esteemed for its digestive promotion, vitality enhancement, and efficacy in treating diseases such as asthma and digestive disorders [2]. In recent years, the cultivation area of *D. polystachya* Turczaninow has expanded significantly due to the vast market demand. To facilitate standardized and refined cultivation management, it is essential to conduct on-site surveys of its distribution. However, the cultivation of *D. polystachya* Turczaninow is mainly undertaken by individual agricultural households. This situation resulted in a fragmented distribution, which heightened the challenges of traditional surveys. Therefore, it is necessary to develop a rapid, accurate, and large-scale approach for *D. polystachya* Turczaninow monitoring.

Remote sensing technology possesses the capability for rapid and repeated observations over regional-scale areas [3]. It offers a faster and more cost-effective method for *D. polystachya* Turczaninow classification. Spectral and temporal features are major characteristics used for crop classification. There are two strategies to use these features. One uses spectral information from monotemporal remote sensing data within a growing season, which is based on the fact that spectral features are the foundation for distinguishing different vegetation [4]. The variation of canopy morphology, biophysical structure, and biochemical compositions often lead to changes in reflection attributes. Therefore, different crops usually exhibit distinct responses across the visible to shortwave infrared spectral. However, some crops show similar reflectance during peak-growing seasons, which increases the difficulty in crop classification [5]. Therefore, the other strategy is combining both spectral and temporal features to describe the reflectance and phenological changes of crops [6]. By considering multitemporal observations, the classification model can learn and adapt to the dynamic nature of crops.

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This is particularly beneficial for capturing subtle changes and nuances in crop features, which leads to improved accuracy compared to monotemporal data.

In recent years, the development of fine-resolution remote sensing has provided an abundance of high-quality data. Therefore, spatial correlation becomes another crucial characteristic for crop classification [7]. Texture is the commonly used spatial feature in previous studies. In contrast to only using spectral features, combining spectral and texture information generally leads to a 5%–10% improvement in classification accuracy [8], [9]. However, due to various factors, such as the size of sliding window, determining suitable texture features can be challenging in specific research. Recently, deep learning algorithms have been widely used for the classification task, which learns feature representations in an end-to-end regime instead of manual feature engineering based on human experience and prior knowledge [10]. Among numerous deep learning algorithms, convolutional neural networks (CNNs) are regarded as one of the most popular and successful approaches used in crop classification. The shared weights in convolutional layers contribute to the extraction of local features, which allows the network to capture intricate spatial details [11]. Moreover, the adeptness of CNNs at generalizing across diverse conditions enhances the reliability of crop classification models. CNNs are frequently used for classification using monotemporal data. Recently, researchers have introduced 1-D CNN, 3-D CNN, and long short-term memory-CNN (LSTM-CNN) models to tackle classification tasks in multitemporal data [12], [13].

Although several studies have explored various methods for classifying crops using remote sensing data, developing an accurate *D. polystachya* Turczaninow classification model remains challenging. One primary limitation is the insufficient quantity of optical remote sensing data obtained during its crucial growth stages, typically from June to August. Therefore, the excessive temporal data during the germination or maturity phases may diminish the significance of essential temporal phases in the classification model. Furthermore, trellising *D. polystachya* Turczaninow plants during the stem-developing stage introduces a distinctive 3-D planting structure. This alteration in the canopy spectral reflectance of *D. polystachya* Turczaninow results in changes in the sensitive bands for classification. However, traditional time series data construction methods usually rely on a single specific vegetation index (VI), which fails to capture this unique change adequately. Moreover, trellising cultivation also impacts the spatial characteristics of *D. polystachya* Turczaninow. However, due to the small area and fragmented distribution of *D. polystachya* Turczaninow plots, it becomes challenging to describe the spatial features adequately using traditional texture features.

To address these challenges, this study proposes a spectral–temporal–spatial feature optimization method to enhance the classification accuracy of *D. polystachya* Turczaninow. Three specific subobjectives are discussed in this article. First, the optimal growth periods for the *D. polystachya* Turczaninow classification are identified. Second, a multi-VI time series data approach is developed to effectively characterize the growth changes of *D. polystachya* Turczaninow, which aims to overcome the limitations of using a single vegetation index. Finally, the spatial feature extraction and representation capabilities of CNNs are assessed on small-scale *D. polystachya* Turczaninow plots. This evaluation is conducted to determine the efficacy of CNNs in capturing the spatial characteristics of *D. polystachya* Turczaninow in a fragmented landscape.

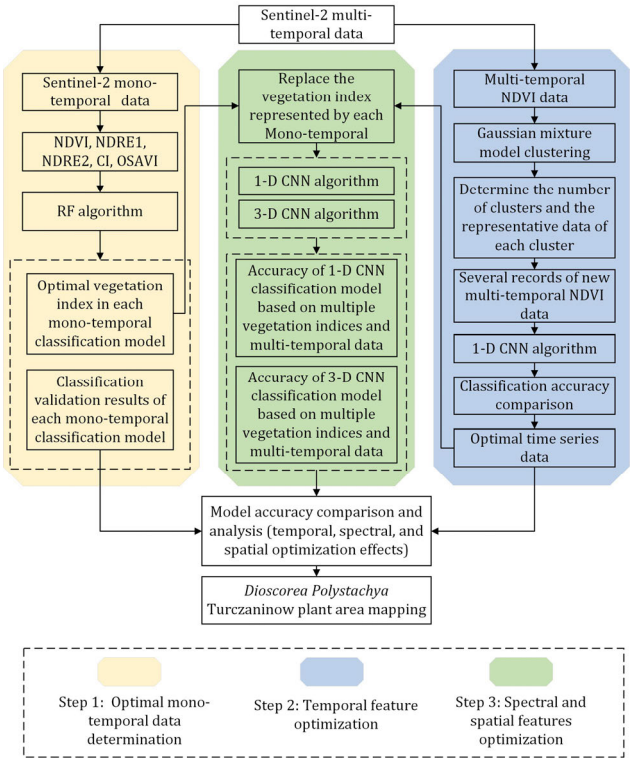



Fig. 1. Flowchart of this study.

## II. Materials and Methods

Fig. 1 shows the flowchart of this study. First, the monotemporal classification model was constructed using the random forest (RF) algorithm and Sentinel-2 data. Five Vis, including normalized difference vegetation index (NDVI), normalized difference red-edge (RE) vegetation index 1 (NDRE1), normalized difference RE vegetation index 2 (NDRE2), chlorophyll index (CI), and optimized soil adjusted vegetation index (OSAVI), were used as inputs for RF model. The classification accuracy and optimal VI of each monotemporal model were used as the evaluation indicators for the temporal and spectral feature optimization process, respectively. Then, the NDVI time series data were constructed using the multitemporal Sentinel-2 data, and the Gaussian mixture model (GMM) algorithm was applied for temporal feature dimensionality reduction. Several reduced NDVI time series data were used as the input of the 1-D CNN classification model, and the optimal time series data were determined through accuracy comparison. Furthermore, the NDVI data were replaced with the optimal VI corresponding to each time phase, thereby obtaining the spectral optimized time series data for *D. polystachya* Turczaninow classification. Next, the temporal and spectral optimized data were used as input for 1-D CNN and 3-D CNN model construction. Finally, the

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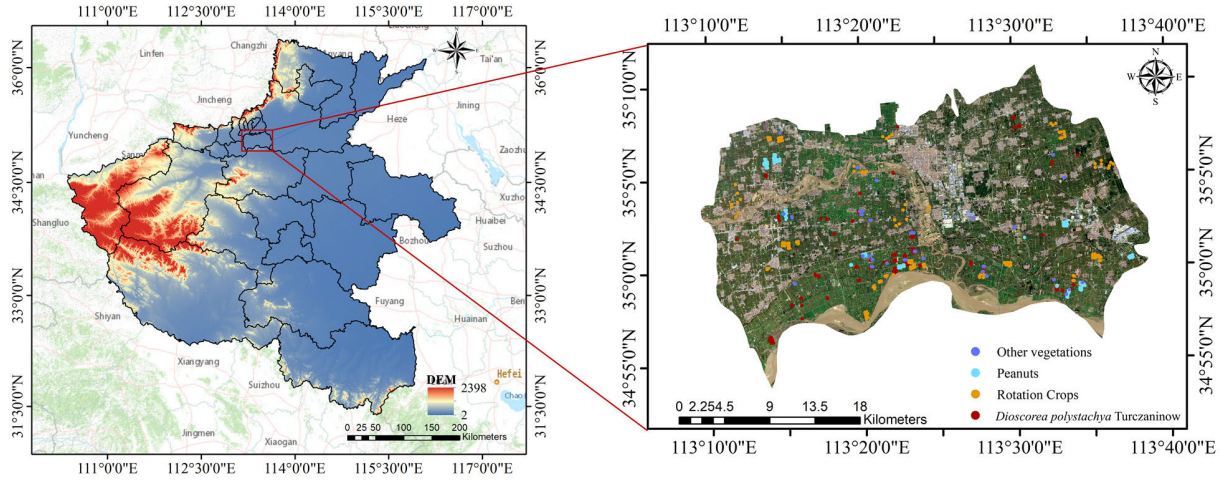



Fig. 2. Location of the study area.

effects of temporal, spectral, and spatial optimization on model accuracy were discussed. The optimal classification model was applied for regional mapping to obtain the spatial distribution of *D. polystachya* Turczaninow cultivation areas in Wuzhi County of China.

### A. Study Area

The study area is located in Wuzhi County (34°56′N35°10′N, 113°10′E113°39′E) of Henan Province, China (Fig. 2). It experiences a temperate continental monsoon climate with an annual average precipitation of 575.1 mm and an average annual temperature of 14.4 °C. Wuzhi County boasts a vast alluvial plain due to its strategic location at the confluence of the Yellow River and Qin River. Consequently, the primary soil type in this county is flavor aquic soil, covering 84.1% of the total area. Wuzhi County is recognized as a pivotal locale in the genesis of various traditional Chinese medicinal plants, notably *D. polystachya* Turczaninow, *Rehmannia glutinosa* (Gaert.) Libosch. ex Fisch. et Mey., *Chrysanthemum morifolium* Ramat., and *Achyranthes bidentata* Blume. Conforming to the agricultural landscape of other central plain regions in China, the county predominantly dedicates its cultivation efforts to winter wheat, summer maize, and peanut, establishing them as primary grain crops. Within the confines of this study area, precisely 11 309.6 hectares of forests. These arboreal expanses play a critical role, serving as indispensable reservoirs to produce timber and other native forest products, thereby contributing significantly to the sustenance of Wuzhi County and its contiguous regions.

### B. Data and Preprocessing

*1) Sentinel-2 Time Series Data and Preprocessing:* Sentinel-2 contains two satellites (S-2A and S-2B) phased at 180° to each other. These satellites provide multispectral data with resolutions of 10–60 m in 13 spectral bands [14]. In this study, two tiles (T49SFU and T49SGU) were selected to cover the area of Wuzhi County. According to the growth

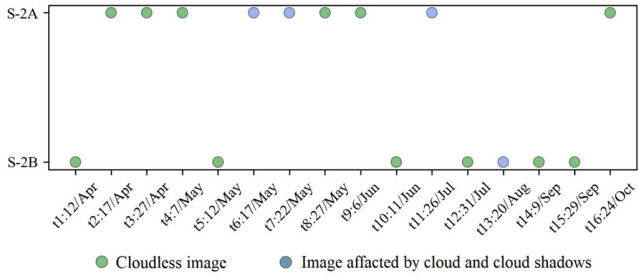



Fig. 3. Sentinel-2 time series data used in this study.

cycle of *D. polystachya* Turczaninow, a total of 16 Sentitnle-2 Level-2A products (t1 to t16) with cloud cover less than 20% were selected in this study (Fig. 3). The time series data span the growth period from sowing (in April) to harvest (in October) in 2021. As shown in Fig. 3, t1–t16 are the Sentinel-2 observation time. Specifically, t1–t3 represent the germination stage, t4–t10 correspond to the stem development stage, and t11–t16 cover the rhizome enlargement stage. To mitigate the impact of Sentinel-2 data quality on classification, a series of preprocessing steps were conducted. First, all bands of each Sentinel-2 data were resampled to 10-m resolution using the Sentinel Application Platform (SNAP) toolbox. Next, the improved Fmask algorithm (3.2. version) was used to remove clouds and cloud shadows [15]. Finally, the geostatistical neighborhood similar pixel interpolator (GNSPI) was used to reconstruct missing pixels in Sentinel-2 data [16]. For each Sentinel-2 data to be interpolated, the data preceding and following it were used as reference inputs for the GNSPI algorithm (Fig. 3). However, due to data quality issues in t6 and t7, the data from t5 and t8 were used as reference to reconstruct the missing pixels.

*2) Field Survey and Reference Data:* Ground surveys were conducted from June 10 to June 13 and from September 30 to October 4, 2021. During ground surveys, the edge coordinates of the target field are recorded using GPS devices. The distribution of vegetation field sample points is shown in Fig. 2. These points are evenly distributed throughout the

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TABLE I
SENTINEL-2 TRAINING AND TESTING SAMPLES USED IN THIS STUDY

| Class | Training dataset | Testing dataset | Total |
|---|---|---|---|
| Rotation crops | 20555 | 9099 | 29654 |
| *Dioscorea polystachya* Turczaninow | 7984 | 4385 | 12369 |
| Peanut | 6143 | 2466 | 8609 |
| Other vegetations | 11858 | 7027 | 18885 |
| Other categories | 7965 | 2097 | 10062 |
| Water | 7595 | 2826 | 10421 |

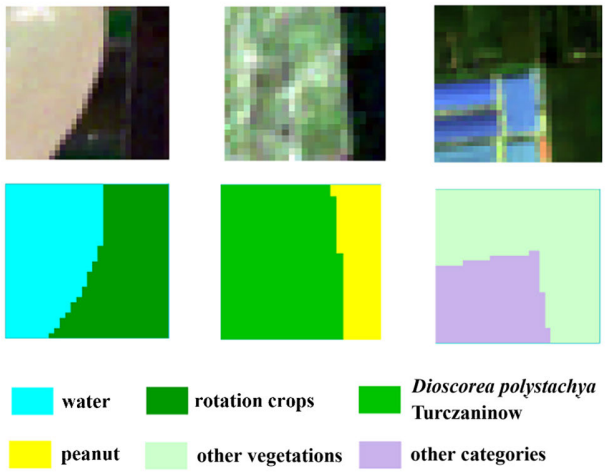



Fig. 4. Representative Sentinel-2 data samples and labels.

entire county. This study used Gaofen-1B panchromatic data with 2-m spatial resolution and Google Earth high-resolution images to generate more reference samples. All samples were divided into six classes: rotation crops, peanut, *D. polystachya* Turczaninow, other vegetations, water, and other categories. Rotation crops contain two planting patterns, including wheat–corn and wheat–peanut. Other vegetations include vegetable planting areas, forests, shrubs, grasslands, and other traditional Chinese medicines. Other categories include impervious surfaces, artificial structures, and soil. Based on ground survey samples and reference samples, 100 image patches with 30 × 30 pixels were selected to generate the training and testing datasets (Fig. 4). Table I shows the number of samples of each category. The total sample size is 90 000 with 62 100 samples in the training dataset (69 image patches) and 27 900 samples in the test dataset (31 image patches).

### C. Monotemporal Classification Model Construction

*1) Vegetation Indices:* Table II shows five VIs used for developing monotemporal classification models. Among those VIs, NDVI is an important index used to reflect the growth and nutritional information of plants. RE is particularly useful in describing plant pigment status and distinguishing between different types of vegetation [17]. In this study, two normalized difference RE VIs were used. Moreover, CI is a ratio vegetation index highly sensitive to chlorophyll content. Since vegetation growth often correlates with changes in canopy chlorophyll content, CI is also an important input for vegetation classification [18]. The OSAVI is an improvement on NDVI. It is less sensitive to soil brightness variations due to introduction of the soil adjustment factor and the optimization constant [19].

*2) RF Classification Algorithm:* RF algorithm was applied to develop monotemporal classification models in this study. RF combines multiple decision trees, each trained on a subset of the data with bootstrapping. During the training process, randomness is introduced by selecting a subset of features for each tree. The final prediction is made through a majority vote of individual trees for classification tasks. This approach increases accuracy and minimizes overfitting. RF is known for its robustness, versatility, and effectiveness in handling various types of datasets, making it a popular choice for classification in data science and machine learning [25]. To improve model performance, this study used a grid search method to determine the number of feature sets (*mtry*) and the number of decision trees (*ntree*). Corresponding to the observed monotemporal Sentinel-2 data, 16 RF classification models (denoted as T1–T16) were developed in this study.

### D. Feature Optimization and Classification Model Construction

To emphasize the feature distinctions between *D. polystachya* Turczaninow and other crops, this study introduced a spectral–temporal–spatial feature optimization framework before developing the classification model. The algorithms employed in each process are described as follows.

*1) Temporal Feature Optimization:* In this study, the original time series data consisted of 16 monotemporal Sentinel-2 NDVI data. The temporal feature optimization process was conducted by eliminating redundant time phases using a clustering algorithm. Considering the temporal span (seven months) and the quantity of images (16 monotemporal data), the number of clusters was set from 7 to 15. Within each cluster, the monotemporal Sentinel-2 data with the highest *D. polystachya* Turczaninow $F1$ score were selected to construct a new time series data. Consequently, a total of nine NDVI time series data were constructed. Afterward, each NDVI time series data was used as the input to develop the classification model. The optimal time phase combination was determined by comparing the $F1$ scores of *D. polystachya* Turczaninow.

In this study, the GMM algorithm was used for time phase clustering. It is an unsupervised algorithm that contains finite Gaussian probability density functions. Therefore, GMM can describe the probability distribution of observation data. The purpose of GMM clustering is to maximize the probability density of the samples [26]. Compared to traditional clustering methods (e.g., $K$-means), GMM clustering can fit any type of data distribution by dividing it into several standard Gaussian models. Therefore, this algorithm is more robust and flexible. To enhance computational efficiency, this study randomly extracted 2000 samples from each category to generate a refined dataset for clustering [27].

The probability density function of GMM is

$$P_M(x) = \sum_{k=1}^{K} p(k)p(x \mid k) = \sum_{k=1}^{K} \alpha_k p\left(x|\mu_k, \sum k\right) \quad (1)$$

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TABLE II
VIs USED FOR DEVELOPING SENTINEL-2 TIME SERIES DATA

| Index | Formula | Characteristic represented by the index | Reference |
|---|---|---|---|
| Normalized difference vegetation index (NDVI) | (NIR1-Red)/(NIR1+Red) | Growth and nutritional information | [20] |
| Normalized difference red-edge vegetation index 1 (NDRE1) | (NIR1-RE1)/(NIR1+RE1) | Physiological status and biomass | [21-22] |
| Normalized difference red-edge vegetation index 2 (NDRE2) | (NIR1-RE2)/(NIR1+RE2) | Physiological status and biomass | [21-22] |
| Chlorophyll index (CI) | (NIR1/Green)-1 | Chlorophyll content status | [23] |
| Optimized soil adjusted vegetation index (OSAVI) | [1.16× (NIR1-Red)]/(NIR1+Red+0.16) | Growth and physiological status | [24] |

TABLE III
STRUCTURE OF 1-D CNN CLASSIFICATION MODEL USED IN THIS STUDY

| Layers | Parameters of 1-D CNN | | |
|---|---|---|---|
| | Channel | Kernel size | Activation function |
| Conv1 | 16 | 5 | ReLU |
| Conv2 | 16 | 5 | ReLU |
| BN1 | | Default | |
| Maxpooling1 | - | 2 | - |
| Conv3 | 32 | 5 | ReLU |
| Conv4 | 32 | 5 | ReLU |
| BN2 | | Default | |
| Maxpooling2 | - | 2 | - |
| Conv5 | 64 | 5 | ReLU |
| Conv6 | 64 | 5 | ReLU |
| BN3 | | Default | |
| Maxpooling3 | - | 2 | - |
| Dropout | | 0.5 | |
| FC | | 6 | |

Notes: Conv stands for the convolutional layer; BN denotes the batch normalization layer; Maxpooling refers to the maximum pooling layer; FC represents the fully connected layer; "-" indicates that the layer does not have that parameter.

where $k$ represents the number of clusters, $\alpha_k$ represents the probability of the $k$th Gaussian distribution, $p(x|k)$ represents probability density of the $k$th Gaussian distribution, $\mu_k$ is the mean vector of $p(x|k)$, and $\sum k$ is the covariance matrix of $p(x|k)$.

During the temporal feature optimization process, the 1-D CNN algorithm was used to develop the classification model. This algorithm uses 1-D filters to capture the spectral patterns inherent in the multitemporal Sentinel-2 data. Within the 1-D CNN model, a pixel is extracted from the entire image cube. The features of pixel are encoded through a multilayer CNN, achieving a label that indicates its respective category. To expedite training convergence and enhance model stability, batch normalization (BN) layer and dropout layers were incorporated into the networks. After multiple parameter adjustments, the structure of the 1-D CNN model is shown in Table III.

*2) Spectral Feature Optimization:* Traditionally, remote sensing temporal data are developed by using the same VI. However, previous studies have demonstrated that the sensitive bands for crops change with their growth phase [28]. In the case of *D. polystachya* Turczaninow, the practice of trellising during the stem-developing stage has altered the attribute of canopy spectral reflectance. Therefore, the sensitive VI for the classification of *D. polystachya* Turczaninow in different monotemporal Sentinel-2 data has also changed. To emphasize this characteristic, this study conducted the spectral feature optimization process by developing multi-VI-based temporal data using the sensitive VI corresponding to each time phase. To identify the most sensitive VI for classification in each monotemporal Sentinel-2 data, two algorithms were used to enhance the robustness of the decision-making process. The RF and Shapley additive explanations (SHAP) algorithms were applied to evaluate the importance and contribution of each VI in the monotemporal classification model. These algorithms are based on different theoretical frameworks. RF is a powerful ensemble learning algorithm that is capable of feature ranking. Based on the Gini index or entropy, the RF algorithm assesses feature importance by measuring its contribution to impurity reduction at each split in the decision trees. The total impurity reduction across all trees is summed and normalized into a global importance score. Features with greater contributions to splitting and impurity reduction receive higher scores. This process reveals the significance of each feature in predicting the class of the target, enabling efficient feature ranking for enhanced model interpretability [29].

Calculating the importance of features in RF is tied to the model structure and the training process. The SHAP algorithm was also used further to enhance the rationality of the VI selection process. SHAP provides a unified framework for interpreting machine learning models based on cooperative game theory. It quantifies the contribution of each feature by evaluating all possible feature subsets and assessing the incremental impact of adding a given feature on model performance. In contrast to the RF algorithm, SHAP is applicable to a broad range of machine learning models, making it a versatile tool for feature importance analysis. This flexibility enables SHAP to provide consistent and comparable interpretations across different types, which is particularly valuable in complex scenarios. In this study, a kernel-based classification algorithm [support vector machines (SVM)] was

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used to combine with SHAP to determine the contribution of each VI. The methodology for calculating SHAP values is outlined as follows [30], [31]:

First, the structure of SVM model needs to be determined to generate the baseline prediction. For each feature $x_i$, all possible subsets of the remaining features were generated to calculate the SHAP value. The SHAP value of $x_i$ with respect to the classification result is computed as the average difference in model performance when the feature is included versus excluded, as described by the following equation:

$$\emptyset_i(f) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (2)$$

where $f(S)$ represents the model performance using the subset $S$; $f(S \cup \{i\})$ represents the model performance when adding $x_i$ to subset $S$; and $|S|$ and $|N|$ represent the number of features in subset $|S|$ and the total number of features, respectively.

To reduce the computational cost during the SHAP value calculation, an approximation technique was employed. In this study, kernel SHAP was used as the approximation method. Finally, the contribution of each feature was determined by calculating the ratio of the absolute SHAP value of feature $x_i$ to the sum of the absolute SHAP values of all features.

To assess and compare the spectral separability among different classes before and after spectral feature optimization, the dynamic time warping (DTW) algorithm was applied to calculate the distances between different time series data. DTW uses dynamic programming to compute the minimum cumulative distance between two time series. The basic idea of DTW is to construct a matrix where each cell represents the cumulative distance between two points in the sequences [32]. The optimal alignment path is identified by tracing back through the matrix from the bottom-right corner to the top-left corner. The DTW distance is the total accumulated cost of the optimal warping path [33]. This study generated reference time series data for each class using the average value of 200 randomly selected samples from the dataset. DTW distances were then calculated using the reference data. This process was repeated 50 times, and the average DTW distance between two classes was used as the final result for comparison.

*3) Spatial Feature Optimization:* Manual-designed spatial features such as gray-level co-occurrence matrix parameters only capture the low-level information, which is usually derived from domain knowledge or insights about the task. By contrast, CNN has a strong capacity for automatic spatial feature extraction due to its use of convolutional layers, which apply filters to capture local patterns, hierarchical representations, and translational invariance [34]. Thus, the spatial feature optimization was performed using a 3-D CNN model to extract the deep features due to the input comprised of multi-VI-based time series data. This algorithm extends conventional 2-D convolutions into the temporal dimension, facilitating the extraction of multiscale spatial features and optimizing spatial information. As the network deepens, each convolutional layer identifies and abstracts different levels of spatial features, creating a hierarchical representation [35]. Therefore, by considering spatial and temporal relationships in three dimensions (width, height, and depth), 3-D CNN captures volumetric information and enables better representation of spatial features over time. In the 3-D CNN model, a patch defined by a window size of $m \times m \times l$ is employed, centered around the target pixel. Subsequently, multiple layers of convolution operations are executed on this patch. After multiple parameter adjustments, the structure of the 1-D CNN model is shown in Table IV.

TABLE IV
STRUCTURE OF 3-D CNN CLASSIFICATION MODEL USED IN THIS STUDY

| Layers | Parameters of 3-D CNN | | |
|---|---|---|---|
| | Channel | Kernel size | Activation function |
| Conv1 | 8 | 5×5×5 | ReLU |
| Conv2 | 8 | 5×5×5 | ReLU |
| Conv3 | 8 | 5×5×5 | ReLU |
| BN1 | | Default | |
| Maxpooling1 | - | 2×2×2 | - |
| Conv4 | 16 | 5×5×5 | ReLU |
| Conv5 | 16 | 5×5×5 | ReLU |
| Conv6 | 16 | 5×5×5 | ReLU |
| BN2 | | Default | |
| Maxpooling2 | - | 2×2×2 | - |
| Conv7 | 32 | 5×5×5 | ReLU |
| Conv8 | 32 | 5×5×5 | ReLU |
| Conv9 | 32 | 5×5×5 | ReLU |
| BN3 | | Default | |
| Maxpooling3 | - | 2×2×2 | - |
| Conv10 | 64 | 5×5×5 | ReLU |
| Conv11 | 64 | 5×5×5 | ReLU |
| Conv12 | 64 | 5×5×5 | ReLU |
| BN4 | | Default | |
| Maxpooling4 | - | 2×2×2 | - |
| Dropout | | 0.5 | |
| FC | | 6 | |

Notes: Conv stands for the convolutional layer; BN denotes the batch normalization layer; Maxpooling refers to the maximum pooling layer; FC represents the fully connected layer; "-" indicates that the layer does not have that parameter.

### E. Accuracy Assessment

The overall accuracy (OA) and $F1$ score of each category were used as indicators for evaluating the model classification performance

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (3)$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100\% \quad (4)$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100\% \quad (5)$$

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (6)$$

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| Confusion Matrix | | Predicted Labels | | |
|---|---|---|---|---|
| | | Category 1 | Category 2 | Category 3 |
| True Labels | Category 1 | A1 | B1 | C1 |
| | Category 2 | A2 | B2 | C2 |
| | Category 3 | A3 | B3 | C3 |

| Confusion Matrix | | Predicted Labels | | |
|---|---|---|---|---|
| | | Category 1 | Category 2 | Category 3 |
| True Labels | Category 1 | A1/Sum1 | B1/Sum1 | C1/Sum1 |
| | Category 2 | A2/Sum2 | B2/Sum2 | C2/Sum2 |
| | Category 3 | A3/Sum3 | B3/Sum3 | C3/Sum3 |

Note: Sum1=A1+B1+C1; Sum2=A2+B2+C2; Sum3=A3+B3+C3

Fig. 5. Confusion matrix used in this study.

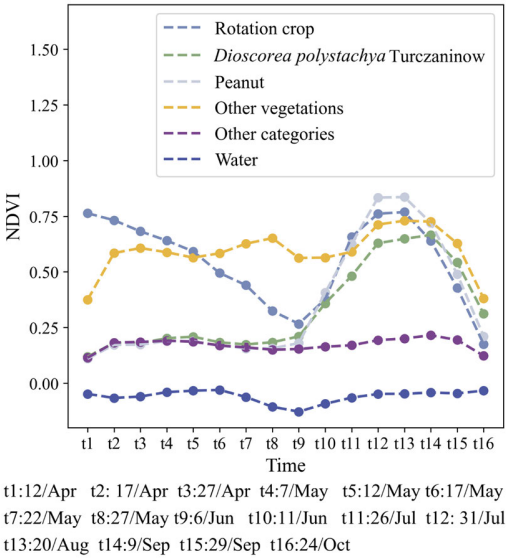



Fig. 6. NDVI time series curve of each category.

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively.

In addition to OA and $F1$ value of each category, this study also employs a confusion matrix to show the classification accuracy of each model. To better represent the *D. polystachya* Turczaninow classification result, this study replaces the numerical values of predicted samples in the confusion matrix with the ratio of predicted sample values to the total number of samples with their true labels (Fig. 5).

## III. Results

### A. Characteristics Analysis of NDVI Time Series Curve in Each Class

The NDVI mean value of all samples in each class is calculated to develop the time series NDVI data (Fig. 6). Within the time range of t1 (April 12) to t16 (October 24), the NDVI curve of rotation crop shows unique changing characteristic with "decreasing–increasing–decreasing" trend. The NDVI curves of *D. polystachya* Turczaninow and peanut show high similarity, especially during the time phase from t1 (April 12) to t10 (June 11). However, the NDVI peak value of *D. polystachya* Turczaninow ranges between 0.6 and 0.7, which is lower than the NDVI peak value of peanut. Additionally, the NDVI peak for *D. polystachya* Turczaninow occurs at t14 (September 9), whereas the peanut NDVI curve reaches its peak at t13 (August 20). Due to the difference in harvesting dates, the NDVI of *D. polystachya* Turczaninow is higher than that of peanut at t16 (October 24). Because other vegetation has many types (such as forests, grasslands, and vegetable planting areas), the trend of this class shows significant fluctuations. By contrast, the NDVI changes in water and other categories show high stability.

### B. Evaluation of Monotemporal Classification Models

Fig. 7 shows the OA values and $F1$ scores for each monotemporal classification model. Among these 16 models, T2 achieved the highest OA at 82.45%. T13 was identified as the most effective monotemporal classification model for *D. polystachya* Turczaninow, with an $F1$ score of 80.39%. Although models T1–T10 demonstrated high OA values (Fig. 7), their capability for *D. polystachya* Turczaninow classification was limited, with $F1$ scores ranging from 60% to 72%. This limitation is attributed to the spectral similarity between *D. polystachya* Turczaninow and peanut during the germination and stem-developing stages (t1 to t10, as shown in Fig. 6), which leads to significant sample confusion. In contrast, models T11–T16 achieved satisfied $F1$ scores for *D. polystachya* Turczaninow, with scores between 77% and 81%. This discrepancy is attributed to the trellising cultivation of *D. polystachya* Turczaninow, which is implemented after mid-May. This treatment not only altered the canopy reflection characteristics of *D. polystachya* Turczaninow but also significantly increased the influence of the underlying surface. Therefore, the changes in the cultivation structure of *D. polystachya* Turczaninow have increased its separability from peanut. However, models T11–T16 exhibited lower OA values ranging from 73% to 80%. This is because the presence of a wheat–peanut planting pattern in rotational crops from t11 to t16 severely affects the classification of peanut and rotational crops. Therefore, these results demonstrate that the rhizome enlargement stage (from t11 to t16) is the optimal growth phase for identifying *D. polystachya* Turczaninow using monotemporal Sentinel-2 data.

### C. Multitemporal Classification Models and Temporal Feature Optimization

This study developed a multitemporal classification model based on 16 Sentinel-2 NDVI data (1-D $CNN_{16t\text{-}NDVI}$). As shown in Fig. 8, the OA and the *D. polystachya* Turczaninow $F1$ score of the 1-D $CNN_{16t\text{-}NDVI}$ model have increased to 85.08% and 83.51%, respectively. Compared to monotemporal classification models, the 1-D $CNN_{16t\text{-}NDVI}$ model achieved minimal variation in $F1$ scores across different classes, indicating that temporal data can enhance the stability and reliability of the model results. Compared to the best monotemporal model for *D. polystachya* Turczaninow

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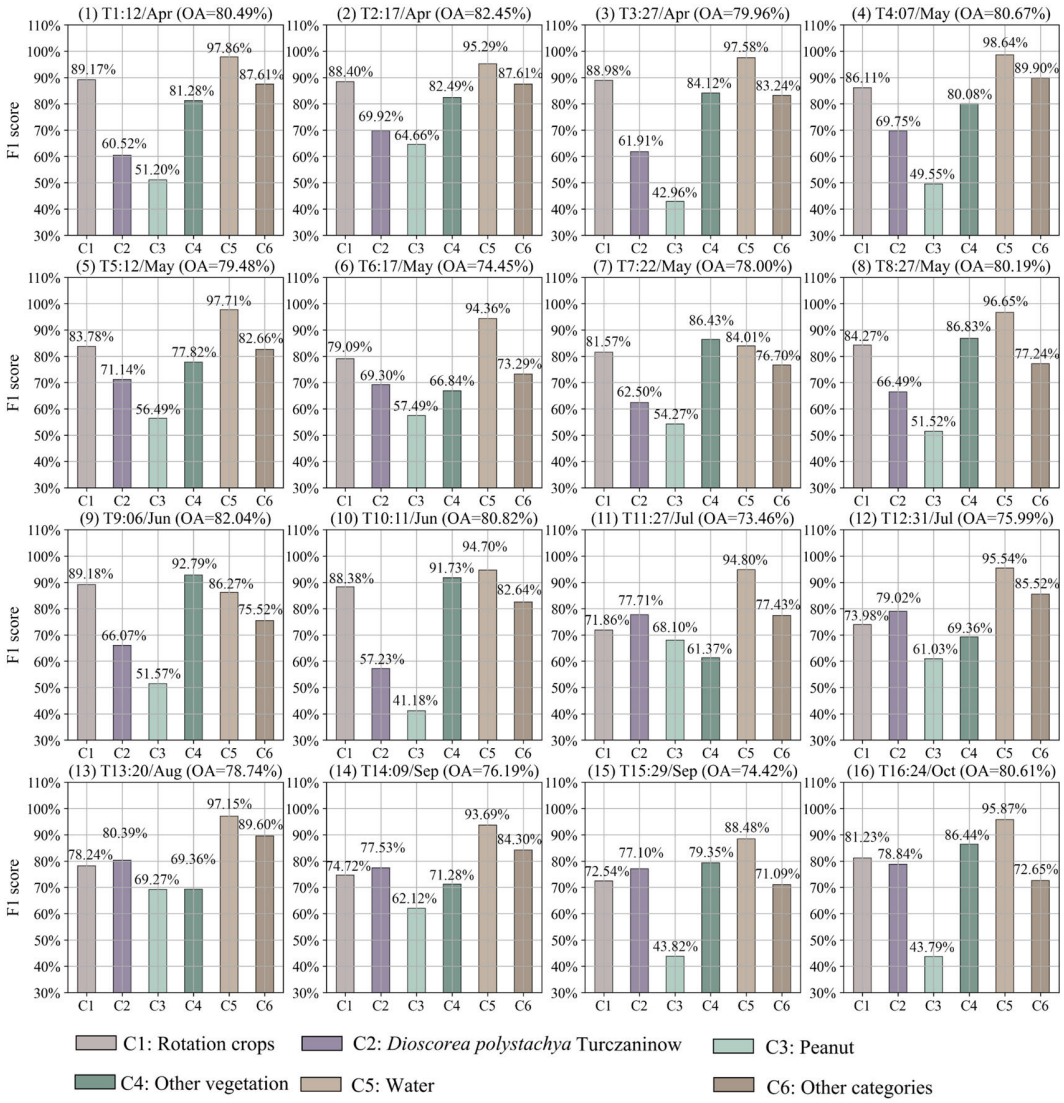



Fig. 7. $F1$ scores and OA values of each monotemporal classification model.

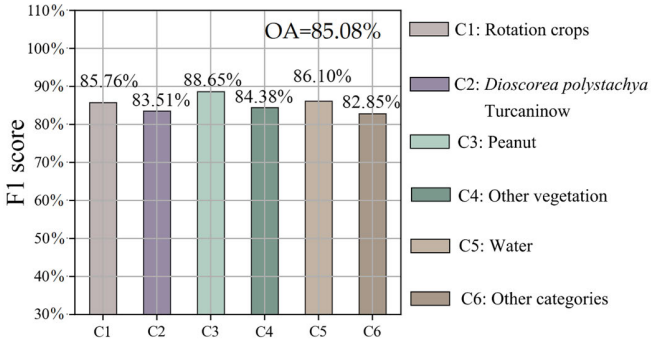



Fig. 8. Evaluation indices of 1-D $CNN_{16t\text{-}NDVI}$ classification model.

classification (T13), the 1-D $CNN_{16t\text{-}NDVI}$ model demonstrates a significant improvement in the classification accuracy of peanut and other vegetations, with $F1$ scores increasing by 19.38% and 15.02%, respectively. However, the $F1$ score for *D. polystachya* Turczaninow only increased by 3.12%. This is because the phenological features represented by multitemporal data significantly enhance the separability between peanut and rotational crops. However, the inputs of 1-D $CNN_{16t\text{-}NDVI}$ model only include R and NIR bands, while model T13 incorporates information from five bands. Consequently, the 1-D $CNN_{16t\text{-}NDVI}$ model utilizes a lower percentage of spectral features, making it less effective in distinguishing between *D. polystachya* Turczaninow and peanut. Therefore, this limitation results in a moderate improvement in identifying *D. polystachya* Turczaninow.

After the GMM clustering process, nine temporal feature-optimized 1-D CNN models were developed. The input temporal data and after temporal optimization model performance of these models are shown in Table V and Fig. 9. Except for M1 and M5, other models show better performance compared to 1-D $CNN_{16t\text{-}NDVI}$ model. M6 achieves the best performance with an OA and $F1$ score improvement for *D. polystachya* Turczaninow of 4.19% and 2.90%, respectively. This improvement is achieved by removing the redundant time phases t1, t3, t5, and t11. Therefore, the input of M6

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TABLE V
SENTINEL-2 TEMPORAL DATA USED IN EACH MODEL

| Model | Input temporal data |
|---|---|
| M1 | t2, t6, t8, t9, t12, t13, t15 |
| M2 | t2, t4, t8, t9, t10, t12, t13, t15 |
| M3 | t2, t4, t7, t8, t9, t10, t12, t13, t15 |
| M4 | t2, t4, t7, t8, t9, t11, t12, t13, t14, t15 |
| M5 | t2, t4, t7, t8, t9, t10, t11, t12, t13, t14, t15 |
| M6 | t2, t4, t6, t7, t8, t9, t10, t12, t13, t14, t15, t16 |
| M7 | t2, t4, t6, t7, t8, t9, t10, t11, t12, t13, t14, t15, t16 |
| M8 | t1, t2, t4, t6, t7, t8, t9, t10, t11, t12, t13, t14, t15, t16 |
| M9 | t1, t2, t3, t4, t6, t7, t8, t9, t10, t11, t12, t13, t14, t15, t16 |

Notes: t1:12/Apr; t2:17/Apr; t3:27/Apr; t4:7/May; t5:12/May; t6:17/May; t7: 22/May; t8:27/May; t9:6/Jun; t10:11/Jun; t11:26/Jul; t12:31/Jul; t13:20/Aug; t14:9/Sep; t15:29/Sep; t16:24/Oct.

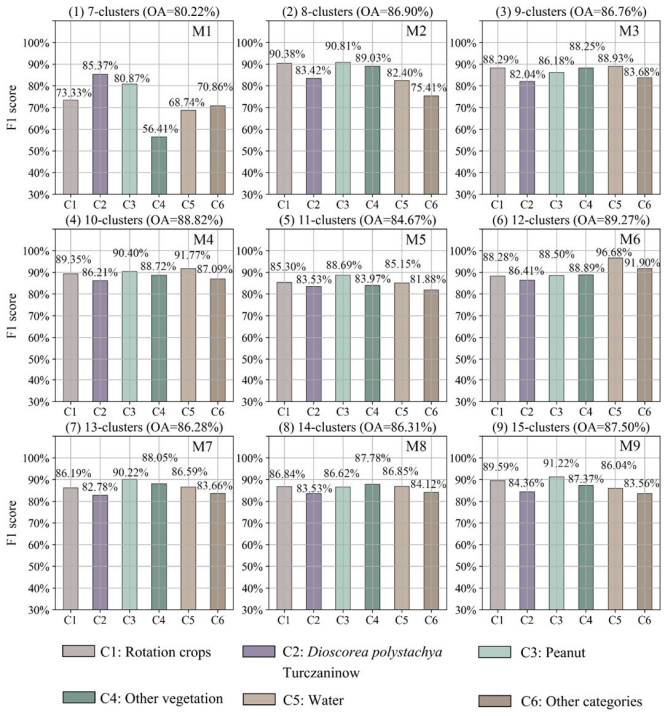



Fig. 9. Evaluation indices of 1-D CNN classification model.

is considered as the optimal temporal data for *D. polystachya* Turczaninow classification.

### *D. Spectral Feature Optimization and Analysis*

Figs. 10 and 11 illustrate the feature importance and SHAP contribution of each VI in the monotemporal classification models. Since the two methods rely on different classification algorithms, the feature rankings vary across some models. However, in each monotemporal classification model, the VI identified as the best by both RF and SHAP is consistent. The optimal VI in each model corresponds to the growth phases of *D. polystachya* Turczaninow. For example, NDRE2 exhibits the highest contribution in T2, which corresponds to

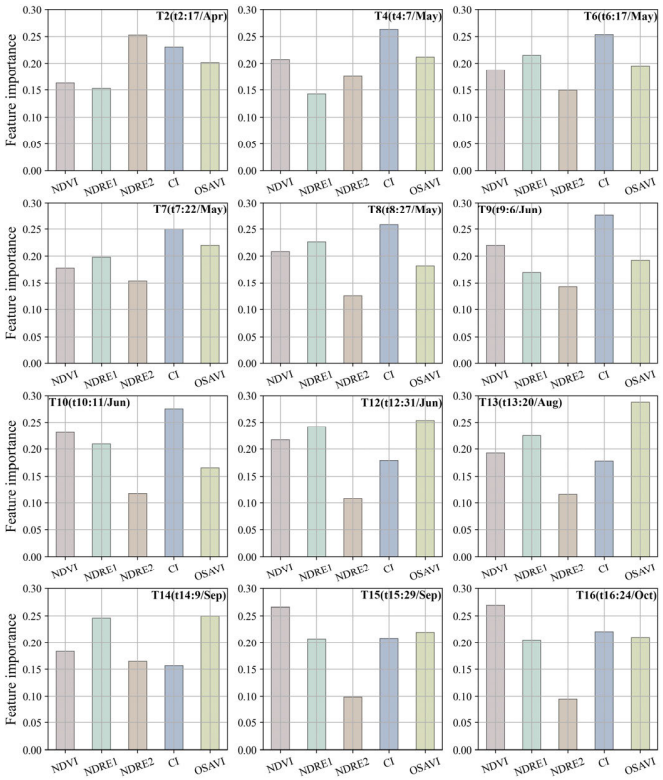



Fig. 10. Feature importance of five VIs in each monotemporal classification mode.

the germination stage. The CI demonstrates the best feature in six monotemporal models (T4 and T6–T10), which represent the stem development stage. The OSAVI and NDVI emerge as the best VIs for four monotemporal models (T13–T16), which corresponds to the rhizome enlargement stage.

After identifying the optimal VIs, this study optimized the spectral features by replacing the original NDVI data with the best VI for each monotemporal classification model. The DTW distances before and after spectral feature optimization are presented in Tables VI and VII. The results show that all DTW distances have increased, which suggests an enhancement in the separability among all classes. For instance, when using the NDVI time series data, the DTW distance between peanut and *D. polystachya Turczaninow* was only 0.7505. This is due to the high similarity in their growth phases and reflectance characteristics. However, after incorporating the multi-VI time series, the DTW distance increased to 2.4271. Moreover, the DTW distance between rotation crops and *D. polystachya Turczaninow* was increased from 2.3122 to 3.9279. Since the class of other vegetation consists of multiple plant types that may exhibit similar temporal trends to the main vegetation types in this study, the spectral separability between these classes is particularly important. As shown in Tables VI and VII, using multi-VI time series data instead of NDVI time series data resulted in an increase of 46.27%, 57.10%, and 34.28% in the DTW distances between other vegetation and rotation crops, *D. polystachya Turczaninow*, and peanut, respectively. These findings demonstrate that using multi-VIs can significantly enhance the separability among different vegetation types.

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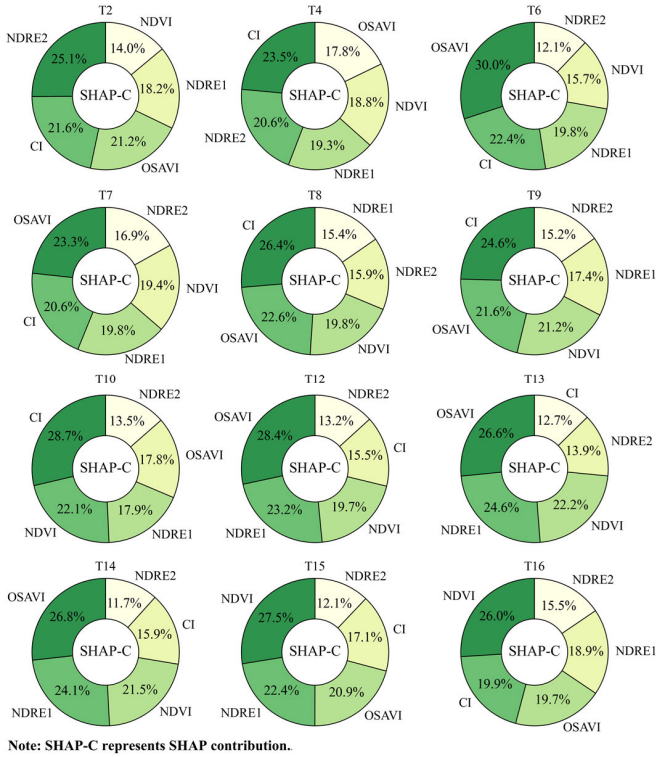



Fig. 11. SHAP contribution of five VIs in each monotemporal classification model.

TABLE VI
DTW DISTANCE BETWEEN SIX CLASSES BASED ON NDVI TIME SERIES DATA

| Class | C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|---|
| C1 | 0 | 2.3112 | 2.2208 | 1.5803 | 3.9483 | 6.7910 |
| C2 | 2.3112 | 0 | 0.7505 | 2.2577 | 3.0889 | 5.0323 |
| C3 | 2.2208 | 0.7505 | 0 | 2.4358 | 3.3735 | 5.2632 |
| C4 | 1.5803 | 2.2577 | 2.4358 | 0 | 5.2391 | 8.0792 |
| C5 | 3.9483 | 3.0889 | 3.3735 | 5.2391 | 0 | 3.8401 |
| C6 | 6.7910 | 5.0323 | 5.2632 | 8.0792 | 3.8401 | 0 |

Notes:C1 represents rotation crops; C2 represents *Dioscorea polystachya* Turczaninow; C3 represents peanut; C4 represents other vegetation; C5 represents other categories; C6 represents water.

### *E. Model Performances After Spectral and Spatial Optimization Processes*

The multi-VI time series data are used as input for developing the 1-D CNN classification model (1-D $CNN_{12t\text{-}multi\text{-}VIs}$). The performance of 1-D $CNN_{12t\text{-}multi\text{-}VIs}$ model is shown in Table VIII and Fig. 12. Compared with M6, the 1-D $CNN_{12t\text{-}multi\text{-}VIs}$ model exhibits an improvement of 2.38% in OA value. The $F1$ scores for rotational crops, *D. polystachya* Turczaninow, peanut, and other vegetation increased by 2.27%, 4.71%, 2.79%, and 1.46%, respectively. As shown in Fig. 12(1) and (2), M6 misclassifies 8.57% of *D. polystachya* Turczaninow samples as rotation crops and 3.11% of rotation crop samples as *D. polystachya* Turczaninow. By contrast, the 1-D $CNN_{12t\text{-}multi\text{-}VIs}$ model reduced these misclassification rates to 4.58% and 2.02%. Additionally, M6 misclassifies 7.10% of peanut and 2.68% of other vegetation as *D. polystachya* Turczaninow, while the 1-D $CNN_{12t\text{-}multi\text{-}VIs}$ model reduces these misclassification rates to 4.30% and 2.26%. Therefore, the optimized multi-VI time series data demonstrate the ability to enhance the differentiation between *D. polystachya* Turczaninow and other classes.

TABLE VII
DTW DISTANCE BETWEEN SIX CLASSES BASED ON MULTI-VI TIME SERIES DATA

| Class | C1 | C2 | C3 | C4 | C5 | C6 |
|---|---|---|---|---|---|---|
| C1 | 0 | 3.9279 | 3.8180 | 2.3115 | 5.9868 | 9.6924 |
| C2 | 3.9279 | 0 | 2.4271 | 3.5468 | 4.3827 | 6.2291 |
| C3 | 3.8180 | 2.4271 | 0 | 3.2708 | 4.8128 | 6.5673 |
| C4 | 2.3115 | 3.5468 | 3.2708 | 0 | 6.7184 | 10.4214 |
| C5 | 5.9868 | 4.3827 | 4.8128 | 6.7184 | 0 | 5.7030 |
| C6 | 9.6924 | 6.2291 | 6.5673 | 10.4214 | 5.7030 | 0 |

Notes:C1 represents rotation crops; C2 represents *Dioscorea polystachya* Turczaninow; C3 represents peanut; C4 represents other vegetation; C5 represents other categories; C6 represents water.

Based on the multi-VI time series data, the 3-D $CNN_{12t\text{-}multi\text{-}VIs}$ model was developed (Table VIII and Fig. 12). By considering the spatial correlation between adjacent pixels in the defined window, this 3-D-structure-based network improved the classification accuracy significantly. The OA of this model has increased to 94.35%. Compared with the 1-D $CNN_{12t\text{-}multi\text{-}VIs}$ model, this model increased the $F1$ scores of *D. polystachya* Turczaninow by 3.88%. Additionally, this model reduced the misclassification rates for rotation crops, peanut, and other vegetation mistakenly identified as *D. polystachya* Turczaninow to 0.74%, 2.47%, and 1.05%, respectively. This result indicated that the 3-D $CNN_{12t\text{-}multi\text{-}VIs}$ model effectively captures the dynamic temporal changes and spatial correlations of various classes. Moreover, to demonstrate the effectiveness of temporal and spectral feature optimization, this study constructed the 3-D $CNN_{16t\text{-}NDVI}$ model based on original NDVI time series data. As shown in Table VIII, the 3-D $CNN_{16t\text{-}NDVI}$ model achieved the OA of 90.55% and $F1$ score of *D. polystachya* Turczaninow of 87.29%. There are 7.18% of *D. polystachya* Turczaninow samples that are identified as rotational crops, and 7.87% of peanut samples that are misclassified as *D. polystachya* Turczaninow (Fig. 12). This result demonstrates that the improvement in *D. polystachya* Turczaninow classification accuracy is limited when using original NDVI time series data. By contrast, the model effectively captures feature differences at different levels after the temporal, spectral, and spatial optimization process. The *D. polystachya* Turczaninow classification results in Wuzhi County are shown in Fig. 13. Most of the planting areas are distributed in the southern (near the Yellow River) and western (near the Qin River) regions. Moreover, the distribution of *D. polystachya* Turczaninow planting fields

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TABLE VIII
PERFORMANCES OF DIFFERENT CLASSIFICATION MODELS

| Model | OA | F1 score | | | | | |
|---|---|---|---|---|---|---|---|
| | | Rotation crops | *Dioscorea polystachya* Turczaninow | Peanut | Other vegetations | Water | Other categories |
| M6 | 89.27% | 88.28% | 86.41% | 88.50% | 88.89% | 96.68% | 91.90% |
| 1-D $CNN_{12t\text{-}multi\text{-}VIs}$ | 91.65% | 90.55% | 91.12% | 91.29% | 90.35% | 97.36% | 94.56% |
| 3-D $CNN_{12t\text{-}multi\text{-}VIs}$ | 94.35% | 93.46% | 95.00% | 94.73% | 92.81% | 98.15% | 96.50% |
| 3-D $CNN_{16t\text{-}NDVI}$ | 90.55% | 90.70% | 87.29% | 91.82% | 91.73% | 92.46% | 89.10% |

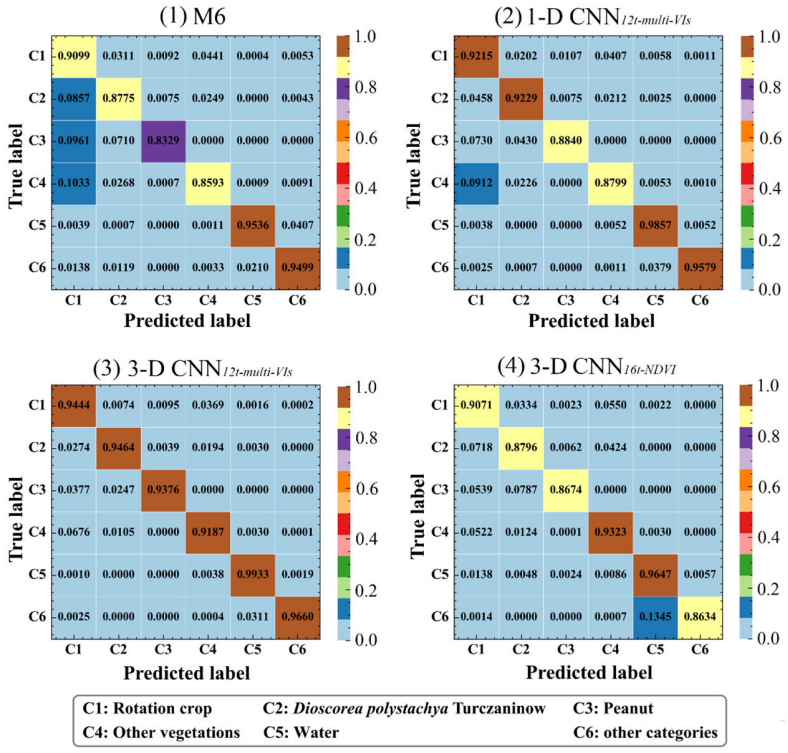



Fig. 12. Confusion matrices for the four classification models.

is fragmented, indicating that self-employed laborers are the primary contributors to the *D. polystachya* Turczaninow cultivation industry in Wuzhi County.

## IV. DISCUSSION

*D. polystachya* Turczaninow demand is driven by a growing interest in traditional medicinal herbs globally. To facilitate efficient resource allocation and production planning, it is imperative to obtain timely and accurate information regarding the area and spatial distribution of *D. polystachya* Turczaninow. However, due to the seasonal variations impacting agricultural land use and land cover [36], the task of *D. polystachya* Turczaninow mapping is still challenging. In such circumstances, optimizing image features based on the characteristics of *D. polystachya* Turczaninow is an effective approach to achieving more reliable results.

Compared to monotemporal classification models, multitemporal classification models enable large-scale dynamic monitoring based on the phenological characteristics exhibited by different crops. In this study, before the temporal feature optimization process, the OA value of the 1-D $CNN_{16\text{-}NDVI}$ model outperforms that of any monotemporal classification model. However, compared with the optimal monotemporal classification model T13, the $F1$ score of the 1-D $CNN_{16\text{-}NDVI}$ model for *D. polystachya* Turczaninow only exhibits a moderate improvement of 3.12%. In addition to the insufficient band information used in the classification model, this result may also be caused by the quantity and quality of Sentinal-2 data. First, using images with high similarity usually introduces significant redundancy in models. As the number of temporal data increases, redundant training samples appear to be enlarged. Introducing more noise from these redundant

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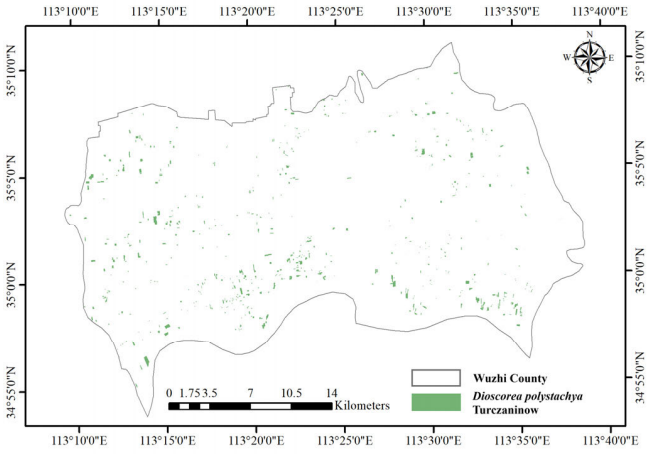



Fig. 13. *D. polystachya* Turczaninow distribution in Wuzhi County.

samples significantly reduces the importance of key growth phases, during which the canopy reflectance of different classes exhibits distinct separability [37], [38]. Therefore, classification accuracy often exhibits an upward trend initially but then stabilizes or even decreases over time [32]. Second, the quality of Sentinal-2 data significantly affects classification accuracy. Although the GNSPI algorithm can increase the available data, the reconstruction errors introduce fluctuations in multitemporal data, which leads to a decline in classification accuracy [34]. For instance, the time phase t11 (26/Jul) is crucial for classification, as both *D. polystachya* Turczaninow and peanut are undergoing a rapid increase in canopy reflectance, with a notable difference in their rate of change. However, the reconstruction errors have introduced reflectance fluctuations into time series data, which reduced the model accuracy (Table V). Therefore, to improve classification accuracy using time series remote sensing data, it is essential to remove the redundant data and ensure the quality of reconstructed data, which are covered by cloud and cloud shadows.

In this study, the temporal optimization indicated that the redundant Sentinel-2 data in the germination and stem-developing stages resulted in a decrease in classification accuracy. This phenomenon can be attributed to the similarity in NDVI values and variations between *D. polystachya* Turczaninow and peanut during these growth stages. As shown in Fig. 6, the NDVI time series data from the early rhizome enlargement stage (t11–t14, from June to September) shows the difference between the two crops, including their respective maximum values and corresponding dates. These differences are attributed to the trellis-style cultivation method and the slow growth characteristics inherent to *D. polystachya* Turczaninow. Thus, this finding suggests that in scenarios characterized by considerable data gaps during critical growth phases, employing the complete dataset spanning the entire growth period is not conducive to enhancing classification accuracy.

Multitemporal classification commonly uses specific VI to characterize the varying attributes of crop canopy reflection. In this study, multi-VI time series data were developed by using sensitive VI in each monotemporal model instead of using NDVI data. The sensitive VI selection result is related to the growth phases of *D. polystachya* Turczaninow. For instance, NDRE2 has higher importance in model T2 because the rotation crop (winter wheat) is in the excessive stage from jointing to heading. The canopy reflectance of the R band is close to saturation during t2 (17/Apr). In contrast, the RE2 band is less affected by the saturation effect, thereby leading to a substantial contribution of NDRE in the T2 model. CI is the sensitive VI in the monotemporal classification model that corresponds to the stem-developing stage. This is because *D. polystachya* Turczaninow completes trellising cultivation in this period. With enhanced light transmission and ventilation, the photosynthetic activity of *D. polystachya* Turczaninow intensifies. In contrast, peanuts exhibit relatively weaker photosynthetic activity, resulting in lower chlorophyll content in the canopy compared to *D. polystachya* Turczaninow. OSAVI and NDVI are the best VIs for classification in the rhizome enlargement stage. This is because OSAVI can mitigate the impact of soil on the canopy reflection characteristics of *D. polystachya* Turczaninow. While during t15 (September 29) and t16 (October 24), rotational crops and peanut have already been harvested, some leaves of *D. polystachya* Turczaninow still remain green. Compared to other RE bands, the R and G bands have higher spatial resolution with 10 m. Therefore, NDVI, CI, and OSAVI all contribute significantly to the T15 and T16 models.

This study conducted a comparative analysis between NDVI and multi-VI time series data (M6 and 1-D $CNN_{12t\text{-}multi\text{-}VIs}$). The result illustrates that using multi-VI time series data leads to a substantial enhancement in classification accuracy. This finding is consistent with prior research [39], which highlighted that although NDVI time series data enhanced the interpretability of classification, employing time series multiband data leads to superior classification accuracy. Similarly, Meng et al. [34] conducted investigations by employing multitemporal Sentinel-2 NDVI and multitemporal "NDVI + full-band reflectance" as inputs for the 1-D CNN model. The outcomes revealed a notable 7.39% enhancement in OA upon including full-band reflectance. These observed limitations can be ascribed to two primary factors. First, single VI contains a restricted amount of spectral information, limiting its capability to describe canopy reflection characteristics. Second, the sensitivity of vegetation to specific wavelengths is frequently influenced by various factors, including vegetation type and atmospheric conditions, resulting in pronounced regional and temporal variations [3]. For instance, Peña and Brenning [6] suggest that during the initial phases of vegetation growth, the significance of the green band outperforms that of the red band. The findings of this study align with these conclusions. During the stem-developing stage of *D. polystachya* Turczaninow, CI achieved the highest feature importance in six monotemporal classification models. While NDVI only achieved the highest feature importance at the late rhizome enlargement stage. Consequently, exclusive reliance on NDVI time series data may lead to the neglect of other crucially sensitive bands.

Other than temporal and spectral features, the spatial correlation of adjacent pixels is also a crucial character in crop

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classification tasks [40]. Compared with Landsat series data, Sentinal-2 data has a higher spatial resolution that can describe detailed spatial information about crops. Previous studies have indicated that adding texture information can increase accuracy by 10%–15% in land cover classification [41]. The finding of this study aligns with this conclusion. Compared to the performance of the 1-D $CNN_{12\text{-multi-VIs}}$ model, the 3-D $CNN_{12\text{-multi-VIs}}$ model demonstrated improved classification accuracy across each category. The $F1$ score of *D. polystachya* Turczaninow in 3-D $CNN_{12\text{-multi-VIs}}$ model has increased by 3.88%, indicating the effectiveness of spatial information. This is attributed to the spatial self-correlation observed in the distribution of *D. polystachya* Turczaninow. However, incorporating this geographic assumption proves challenging in the 1-D $CNN_{12\text{-multi-VIs}}$ model. Another reason for these accuracy improvements is that the 3-D CNN classification model possesses the capability to achieve multilayered representation through hierarchical extraction of image features. This capability allows this network to model intricate relationships using a gradient descent approach, ultimately facilitating the automatic learning of image features [12], [42]. In contrast, machine learning classification algorithms such as RF require prior knowledge and manual selection [43]. Therefore, using machine learning algorithms for fine crop classification tasks is still challenging.

While the findings of this study represent significant progress, several aspects still require further optimization. For instance, *D. polystachya* Turczaninow is typically cultivated in well-drained, gently sloping areas. Therefore, integrating terrain characteristics, such as slope and elevation, into the classification model offers a promising approach to improve accuracy. Additionally, the results of this study demonstrate that monotemporal Sentinel-2 data collected between July and August show strong potential for identifying *D. polystachya* Turczaninow. The spectral differences between *D. polystachya* Turczaninow and peanut are clearly distinguishable during this specific time window. Moreover, testing the classification performance of *D. polystachya* Turczaninow based on time series Sentinel-2 data across different growth stages is also crucial for developing representative time series data. This approach helps develop more efficient and lightweight classification models. However, cloud cover has severely affected the implementation of the above work. Future research will focus on using spatiotemporal fusion techniques to reconstruct high-quality time series Sentinel-2 data to test the classification performance of monotemporal data and multitemporal data of each growth stage. Furthermore, the distinctive planting structure of *D. polystachya Turczaninow* leads to unique texture features and scattering characteristics. Thus, integrating multisource sensors, such as Gaofen-2 and Sentinel-1, could diversify the feature set and further enhance class separability.

## V. Conclusion

This study proposed a classification method for *D. polystachya* Turczaninow based on optimizing spectral–temporal–spatial features derived from Sentinel-2 time series data. The multi-VI time series data generated in this study provide valuable and distinctive features across different vegetation types, which results in a notable improvement in classification accuracy. The results indicate that applying the 3-D CNN model to multi-VI time series data compensates for the limited spectral information inherent in multispectral data, enabling more accurate large-scale classification of *D. polystachya* Turczaninow. The main conclusions of this study are as follows.

1) The rhizome enlargement stage represents the optimal growth phase for classifying *D. polystachya* Turczaninow using monotemporal data. The best temporal phase occurs at t13 (20/Aug), with an $F1$ score of 80.39%.
2) Data from the germination and rhizome enlargement stages are crucial for improving model accuracy when using Sentinel-2 time series data for classification.
3) The sensitive VIs for classifying *D. polystachya* Turczaninow vary across growth stages. Compared to NDVI, the use of multi-VIs significantly enhances the separability between vegetation types. The optimal multi-VI time series data are constructed using NDRE2, CI, OSAVI, and NDVI.
4) The optimization and integration of spectral–temporal–spatial features effectively capture the temporal dynamics and spatial correlations of *D. polystachya* Turczaninow, contributing to a more robust and accurate classification.

## References

[1] Z. Chen, T. Shi, X. Zhang, K. Jia, H. Jiang, and B. Yuan, "A hybrid leaf area index estimation method of *Dioscorea Polystachya Turczaninow* using Sentinel-2 vegetation indices," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4415713, doi: 10.1109/TGRS.2022.3219981.

[2] J. Epping and N. Laibach, "An underutilized orphan tuber crop—Chinese yam: A review," *Planta*, vol. 252, no. 4, p. 58, Oct. 2020, doi: 10.1007/s00425-020-03458-3.

[3] C. Sun, J. Li, L. Cao, Y. Liu, S. Jin, and B. Zhao, "Evaluation of vegetation index-based curve fitting models for accurate classification of salt Marsh vegetation using Sentinel-2 time-series," *Sensors*, vol. 20, no. 19, p. 5551, Sep. 2020, doi: 10.3390/s20195551.

[4] S. Marino and A. Alvino, "Vegetation indices data clustering for dynamic monitoring and classification of wheat yield crop traits," *Remote Sens.*, vol. 13, no. 4, p. 541, Feb. 2021, doi: 10.3390/rs13040541.

[5] S. Li, L. Xu, Y. Jing, H. Yin, X. Li, and X. Guan, "High-quality vegetation index product generation: A review of NDVI time series reconstruction techniques," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 105, Dec. 2021, Art. no. 102640, doi: 10.1016/j.jag.2021.102640.

[6] M. A. Peña and A. Brenning, "Assessing fruit-tree crop classification from Landsat-8 time series for the Maipo Valley, Chile," *Remote Sens. Environ.*, vol. 171, pp. 234–244, Dec. 2015, doi: 10.1016/j.rse.2015.10.029.

[7] S. Bera and V. K. Shrivastava, "Analysis of various optimizers on deep convolutional neural network model in the application of hyperspectral remote sensing image classification," *Int. J. Remote Sens.*, vol. 41, no. 7, pp. 2664–2683, Apr. 2020, doi: 10.1080/01431161.2019.1694725.

[8] V. N. Mishra, R. Prasad, P. K. Rai, A. K. Vishwakarma, and A. Arora, "Performance evaluation of textural features in improving land use/land cover classification accuracy of heterogeneous landscape using multisensor remote sensing data," *Earth Sci. Informat.*, vol. 12, no. 1, pp. 71–86, Mar. 2019, doi: 10.1007/s12145-018-0369-z.

[9] X. Zhang, L. Han, L. Han, and L. Zhu, "How well do deep learning-based methods for land cover classification and object detection perform on high resolution remote sensing imagery?" *Remote Sens.*, vol. 12, no. 3, p. 417, Jan. 2020, doi: 10.3390/rs12030417.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:52:19 UTC from IEEE Xplore. Restrictions apply.

[10] A. E. Maxwell, T. A. Warner, and L. A. Guillén, "Accuracy assessment in convolutional neural network-based deep learning remote sensing studies—Part 1: Literature review," *Remote Sens.*, vol. 13, no. 13, p. 2450, Jun. 2021, doi: 10.3390/rs13132450.

[11] K. Bhosle and V. Musande, "Evaluation of deep learning CNN model for land use land cover classification and crop identification using hyperspectral remote sensing images," *J. Indian Soc. Remote Sens.*, vol. 47, no. 11, pp. 1949–1958, Nov. 2019, doi: 10.1007/s12524-019-01041-2.

[12] J. Yin, C. Qi, Q. Chen, and J. Qu, "Spatial-spectral network for hyperspectral image classification: A 3-D CNN and bi-LSTM framework," *Remote Sens.*, vol. 13, no. 12, p. 2353, Jun. 2021, doi: 10.3390/rs13122353.

[13] H. Zhao, S. Duan, J. Liu, L. Sun, and L. Reymondin, "Evaluation of five deep learning models for crop type mapping using Sentinel-2 time series images with missing information," *Remote Sens.*, vol. 13, no. 14, p. 2790, Jul. 2021, doi: 10.3390/rs13142790.

[14] M. Drusch et al., "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," *Remote Sens. Environ.*, vol. 120, pp. 25–36, May 2012, doi: 10.1016/j.rse.2011.11.026.

[15] Z. Zhu, S. Wang, and C. E. Woodcock, "Improvement and expansion of the Fmask algorithm: Cloud, cloud shadow, and snow detection for Landsats 4–7, 8, and Sentinel 2 images," *Remote Sens. Environ.*, vol. 159, pp. 269–277, Mar. 2015, doi: 10.1016/j.rse.2014.12.014.

[16] X. Zhu, D. Liu, and J. Chen, "A new geostatistical approach for filling gaps in Landsat ETM+ SLC-off images," *Remote Sens. Environ.*, vol. 124, pp. 49–60, Sep. 2012, doi: 10.1016/j.rse.2012.04.019.

[17] A. M. Abdi, "Land cover and land use classification performance of machine learning algorithms in a boreal landscape using Sentinel-2 data," *GISci. Remote Sens.*, vol. 57, no. 1, pp. 1–20, Jan. 2020, doi: 10.1080/15481603.2019.1650447.

[18] H. Ye et al., "Recognition of banana fusarium wilt based on UAV remote sensing," *Remote Sens.*, vol. 12, no. 6, p. 938, Mar. 2020, doi: 10.3390/rs12060938.

[19] A. Capolupo, C. Monterisi, and E. Tarantino, "Landsat images classification algorithm (LICA) to automatically extract land cover information in Google Earth engine environment," *Remote Sens.*, vol. 12, no. 7, p. 1201, Apr. 2020, doi: 10.3390/rs12071201.

[20] J. W. Rouse, R. H. Haas, J. A. Schell, and D. W. Deering, "Monitoring vegetation systems in the great plains with ERTS," Texas A&M Univ., College Station, TX, USA, Tech. Rep. PAPER-A20, 1974.

[21] A. Gitelson and M. N. Merzlyak, "Spectral reflectance changes associated with autumn senescence of *Aesculus hippocastanum* L. and Acer *platanoides* L. Leaves. spectral features and relation to chlorophyll estimation," *J. Plant Physiol.*, vol. 143, no. 3, pp. 286–292, Mar. 1994, doi: 10.1016/S0176-1617(11)81633-0.

[22] T. Dong et al., "Assessment of red-edge vegetation indices for crop leaf area index estimation," *Remote Sens. Environ.*, vol. 222, pp. 133–143, Mar. 2019, doi: 10.1016/j.rse.2018.12.032.

[23] A. A. Gitelson, Y. Gritz, and M. N. Merzlyak, "Relationships between leaf chlorophyll content and spectral reflectance and algorithms for non-destructive chlorophyll assessment in higher plant leaves," *J. Plant Physiol.*, vol. 160, no. 3, pp. 271–282, Jan. 2003, doi: 10.1078/0176-1617-00887.

[24] A. R. Huete, "A soil-adjusted vegetation index (SAVI)," *Remote Sens. Environ.*, vol. 25, no. 3, pp. 295–309, Aug. 1988, doi: 10.1016/0034-4257(88)90106-X.

[25] M. P. Neto and F. V. Paulovich, "Explainable matrix–visualization for global and local interpretability of random forest classification ensembles," *IEEE Trans. Vis. Comput. Graphics*, vol. 27, no. 2, pp. 1427–1437, Feb. 2021, doi: 10.1109/TVCG.2020.3030354.

[26] J. Shi, Q. He, and Z. Wang, "GMM clustering-based decision trees considering fault rate and cluster validity for analog circuit fault diagnosis," *IEEE Access*, vol. 7, pp. 140637–140650, 2019, doi: 10.1109/ACCESS.2019.2943380.

[27] G. J. McLachlan and S. Rathnayake, "On the number of components in a Gaussian mixture model," *WIREs Data Mining Knowl. Discovery*, vol. 4, no. 5, pp. 341–355, Sep. 2014, doi: 10.1002/widm.1135.

[28] Z. M. Al-Ali, M. M. Abdullah, N. B. Asadalla, and M. Gholoum, "A comparative study of remote sensing classification methods for monitoring and assessing desert vegetation using a UAV-based multispectral sensor," *Environ. Monitor. Assessment*, vol. 192, no. 6, p. 389, Jun. 2020, doi: 10.1007/s10661-020-08330-1.

[29] A. P. M. Ramos et al., "A random forest ranking approach to predict yield in maize with UAV-based vegetation spectral indices," *Comput. Electron. Agricult.*, vol. 178, Nov. 2020, Art. no. 105791, doi: 10.1016/j.compag.2020.105791.

[30] Y. Nohara, K. Matsumoto, H. Soejima, and N. Nakashima, "Explanation of machine learning models using Shapley additive explanation and application for real data in hospital," *Comput. Methods Programs Biomed.*, vol. 214, Feb. 2022, Art. no. 106584, doi: 10.1016/j.cmpb.2021.106584.

[31] L. Antwarg, R. M. Miller, B. Shapira, and L. Rokach, "Explaining anomalies detected by autoencoders using Shapley additive explanations," *Expert Syst. Appl.*, vol. 186, Dec. 2021, Art. no. 115736, doi: 10.1016/j.eswa.2021.115736.

[32] L. Fan et al., "The effects of Landsat image acquisition date on winter wheat classification in the North China plain," *ISPRS J. Photogramm. Remote Sens.*, vol. 187, pp. 1–13, May 2022, doi: 10.1016/j.isprsjprs.2022.02.016.

[33] H. Li, J. Liu, Z. Yang, R. W. Liu, K. Wu, and Y. Wan, "Adaptively constrained dynamic time warping for time series classification and clustering," *Inf. Sci.*, vol. 534, pp. 97–116, Sep. 2020, doi: 10.1016/j.ins.2020.04.009.

[34] S. Meng, X. Wang, X. Hu, C. Luo, and Y. Zhong, "Deep learning-based crop mapping in the cloudy season using one-shot hyperspectral satellite imagery," *Comput. Electron. Agricult.*, vol. 186, Jul. 2021, Art. no. 106188, doi: 10.1016/j.compag.2021.106188.

[35] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2020.3043710.

[36] D. Bargiel, "A new method for crop classification combining time series of radar images and crop phenology information," *Remote Sens. Environ.*, vol. 198, pp. 369–383, Sep. 2017, doi: 10.1016/j.rse.2017.06.022.

[37] Q. Hu et al., "A phenology-based spectral and temporal feature selection method for crop mapping from satellite time series," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 80, pp. 218–229, Aug. 2019, doi: 10.1016/j.jag.2019.04.014.

[38] H. Zhang, J. Kang, X. Xu, and L. Zhang, "Accessing the temporal and spectral features in crop type mapping using multi-temporal Sentinel-2 imagery: A case study of Yi'an county, Heilongjiang province, China," *Comput. Electron. Agricult.*, vol. 176, Sep. 2020, Art. no. 105618, doi: 10.1016/j.compag.2020.105618.

[39] J. P. Guerschman, J. M. Paruelo, C. D. Bella, M. C. Giallorenzi, and F. Pacin, "Land cover classification in the Argentine Pampas using multi-temporal Landsat TM data," *Int. J. Remote Sens.*, vol. 24, no. 17, pp. 3381–3402, Jan. 2003, doi: 10.1080/0143116021000021288.

[40] X.-Y. Tong et al., "Land-cover classification with high-resolution remote sensing images using transferable deep models," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111322, doi: 10.1016/j.rse.2019.111322.

[41] S. E. Franklin, R. J. Hall, L. M. Moskal, A. J. Maudie, and M. B. Lavigne, "Incorporating texture into classification of forest species composition from airborne multispectral images," *Int. J. Remote Sens.*, vol. 21, no. 1, pp. 61–79, Jan. 2000, doi: 10.1080/014311600210993.

[42] X. Huang et al., "A multispectral and multiangle 3-D convolutional neural network for the classification of ZY-3 satellite images over urban areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10266–10285, Dec. 2021, doi: 10.1109/TGRS.2020.3037211.

[43] S. Wan and S.-H. Chang, "Crop classification with WorldView-2 imagery using support vector machine comparing texture analysis approaches and grey relational analysis in Jianan plain, Taiwan," *Int. J. Remote Sens.*, vol. 40, no. 21, pp. 8076–8092, Nov. 2019, doi: 10.1080/01431161.2018.1539275.

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